

AI assistance and provenance

Repository source: AI_ASSISTANCE.md

Named author: Bruce J. Swihart

AI system used: OpenAI ChatGPT (GPT-5.6 Pro)

Access period: August-September 2026

Scope of AI assistance

OpenAI ChatGPT (GPT-5.6 Pro) was used extensively during this project to:

- discuss and compare possible proof strategies;
- derive, reorganize, and check mathematical calculations;
- generate and execute exact symbolic Python checks;
- prepare matching independent R cross-checks;
- draft and revise the mathematical exposition;
- prepare LaTeX, Markdown, build scripts, and continuous-integration files; and
- help organize the reproducibility and verification records.

The AI system was not used as an author and is not presented as an independent reviewer. Its outputs can contain mathematical, computational, or expository errors.

Human contribution and responsibility

Bruce J. Swihart selected the research problem and project goals, supplied and reviewed the source materials, chose which proposed directions to pursue, ran the independent R checks locally, created and maintained the repository, reviewed the manuscript and supporting documents, communicated the work's status to outside researchers, and decides whether and when any version is made public.

The named author is responsible for accurately describing the provenance and verification status of the repository, preserving corrections, and not presenting the draft as peer reviewed or independently verified before such review occurs.

Verification boundary

The repository contains exact symbolic computations, independent Python and R numerical tests, a human-readable derivation, automated clean-environment checks, and an internal AI-assisted proof audit. These layers improve reproducibility and make errors easier to locate, but they do not constitute independent subject-matter review or peer review.

The current public-facing designation is:

AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification.

Model-identification note

The model designation records the system used in this project at the time of access. Hosted AI systems may change over time, so the designation and access period are included for provenance rather than as a claim of reproducibility of model behavior.